

Tamagotchi specification

A Tamagotchi is a digital pet.

The Tamagotchi's internal state is represented by four numbers, each in the range 0–10. These numbers determine how happy, hungry, tired and dirty the Tamagotchi is. The meaning of these numbers is given below:

- The Tamagotchi is “asleep” if its tiredness is 10.
- It is “hungry” if its hunger is at least 7.
- It is “tired” if its tiredness is at least 8.
- It is “dirty” if its dirtiness is at least 7.
- It is “happy” if its happiness is at least 7.
- It is “content” if its happiness is between 3 and 6.
- It is “sad” if its happiness is less than 3.

With a Tamagotchi, you can:

- Feed the Tamagotchi, making it less hungry. This method takes no parameters and returns nothing.
- Take the Tamagotchi for a walk, making it happier but more tired. This method takes no parameters and returns nothing.
- Pet the Tamagotchi, making it happier. This method takes no parameters and returns nothing.
- Clean the Tamagotchi, making it less dirty. This method takes no parameters and returns nothing.
- See the Tamagotchi's current mood, according to the rules above. This method takes no parameters, and returns the mood as a string.
- Leave the Tamagotchi alone for a short while, after which it is slightly hungrier, slightly dirtier, and slightly more tired. If the Tamagotchi is asleep then it wakes up, having a tiredness of 0. The Tamagotchi also becomes less happy if it is dirty or hungry. This method takes no parameters and returns nothing.

After each method, if any of the happiness, hungriness, tiredness or dirtiness numbers are below 0, they should be reset to 0; if any are above 10, they should be reset to 10.